

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

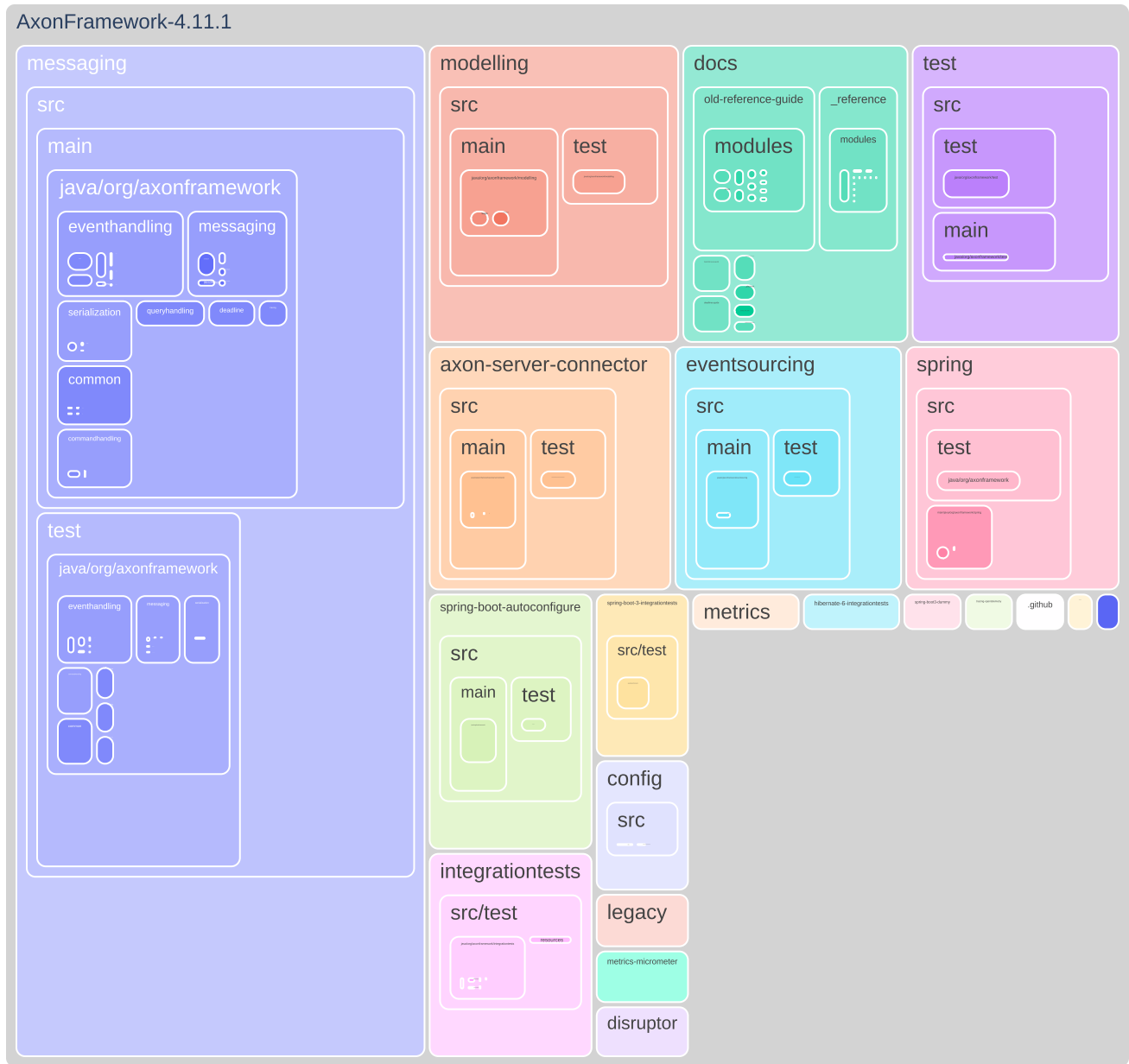
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	563.000000	563.000000	563.000000	563.000000	563.000000	563.000000
mean	24.492007	19.502664	1596.113677	206.362345	1063.900533	299.301954
std	117.928146	21.209040	7954.335528	127.121149	968.697642	367.525167
min	1.000000	2.000000	3.000000	59.000000	65.000000	58.000000
25%	1.000000	7.000000	50.500000	123.000000	305.000000	122.000000
50%	4.000000	12.000000	141.000000	265.000000	824.000000	264.000000
75%	10.000000	25.000000	595.500000	265.000000	1878.000000	264.000000
max	2250.000000	212.000000	154891.000000	2415.000000	5542.000000	2893.000000

	directoryPath	directoryParentPat
0	AxonFramework-4.11.1/assembly	AxonFramework-4.11.
1	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source	AxonFramework-4.11.1/axon-serve
2	AxonFramework-4.11.1/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-4.11.1/axon-server-connector/src/main/resource
3	AxonFramework-4.11.1/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util	AxonFramework-4.11.1/axon-serve
4	AxonFramework-4.11.1/config/src/test/resources/META-INF	AxonFramework-4.11.1/config/src/test/resource
5	AxonFramework-4.11.1/coverage-report	AxonFramework-4.11.
6	AxonFramework-4.11.1/docs/_playbook/localLinks	AxonFramework-4.11.1/docs/_playbo
7	AxonFramework-4.11.1/docs/_reference/modules/ROOT/attachments	AxonFramework-4.11.1/docs/_reference/modules/ROO
8	AxonFramework-4.11.1/docs/_reference/modules/ROOT/partials	AxonFramework-4.11.1/docs/_reference/modules/ROO
9	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connect
10	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connect
11	AxonFramework-4.11.1/docs/_reference/modules/configuration/pages	AxonFramework-4.11.1/docs/_reference/modules/configuratio
12	AxonFramework-4.11.1/docs/_reference/modules/configuration/partials	AxonFramework-4.11.1/docs/_reference/modules/configuratio
13	AxonFramework-4.11.1/docs/_reference/modules/disruptor/pages	AxonFramework-4.11.1/docs/_reference/modules/disruptor
14	AxonFramework-4.11.1/docs/_reference/modules/disruptor/partials	AxonFramework-4.11.1/docs/_reference/modules/disruptor
15	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.11.1/docs/_reference/modules/eventssourcin
16	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.11.1/docs/_reference/modules/eventssourcin
17	AxonFramework-4.11.1/docs/_reference/modules/legacy/pages	AxonFramework-4.11.1/docs/_reference/modules/legac
18	AxonFramework-4.11.1/docs/_reference/modules/legacy/partials	AxonFramework-4.11.1/docs/_reference/modules/legac
19	AxonFramework-4.11.1/docs/_reference/modules/metrics/pages	AxonFramework-4.11.1/docs/_reference/modules/metric
20	AxonFramework-4.11.1/docs/_reference/modules/metrics/partials	AxonFramework-4.11.1/docs/_reference/modules/metric
21	AxonFramework-4.11.1/docs/_reference/modules/micrometer/pages	AxonFramework-4.11.1/docs/_reference/modules/micromete
22	AxonFramework-4.11.1/docs/_reference/modules/micrometer/partials	AxonFramework-4.11.1/docs/_reference/modules/micromete
23	AxonFramework-4.11.1/docs/_reference/modules/modeling/partials	AxonFramework-4.11.1/docs/_reference/modules/modelin
24	AxonFramework-4.11.1/docs/_reference/modules/spring/pages	AxonFramework-4.11.1/docs/_reference/modules/sprin
25	AxonFramework-4.11.1/docs/_reference/modules/spring/partials	AxonFramework-4.11.1/docs/_reference/modules/sprin
26	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.11.1/docs/_reference/modules/springboot-start
27	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.11.1/docs/_reference/modules/springboot-start
28	AxonFramework-4.11.1/docs/_reference/modules/test/pages	AxonFramework-4.11.1/docs/_reference/modules/te
29	AxonFramework-4.11.1/docs/_reference/modules/test/partials	AxonFramework-4.11.1/docs/_reference/modules/te

Number of files per directory

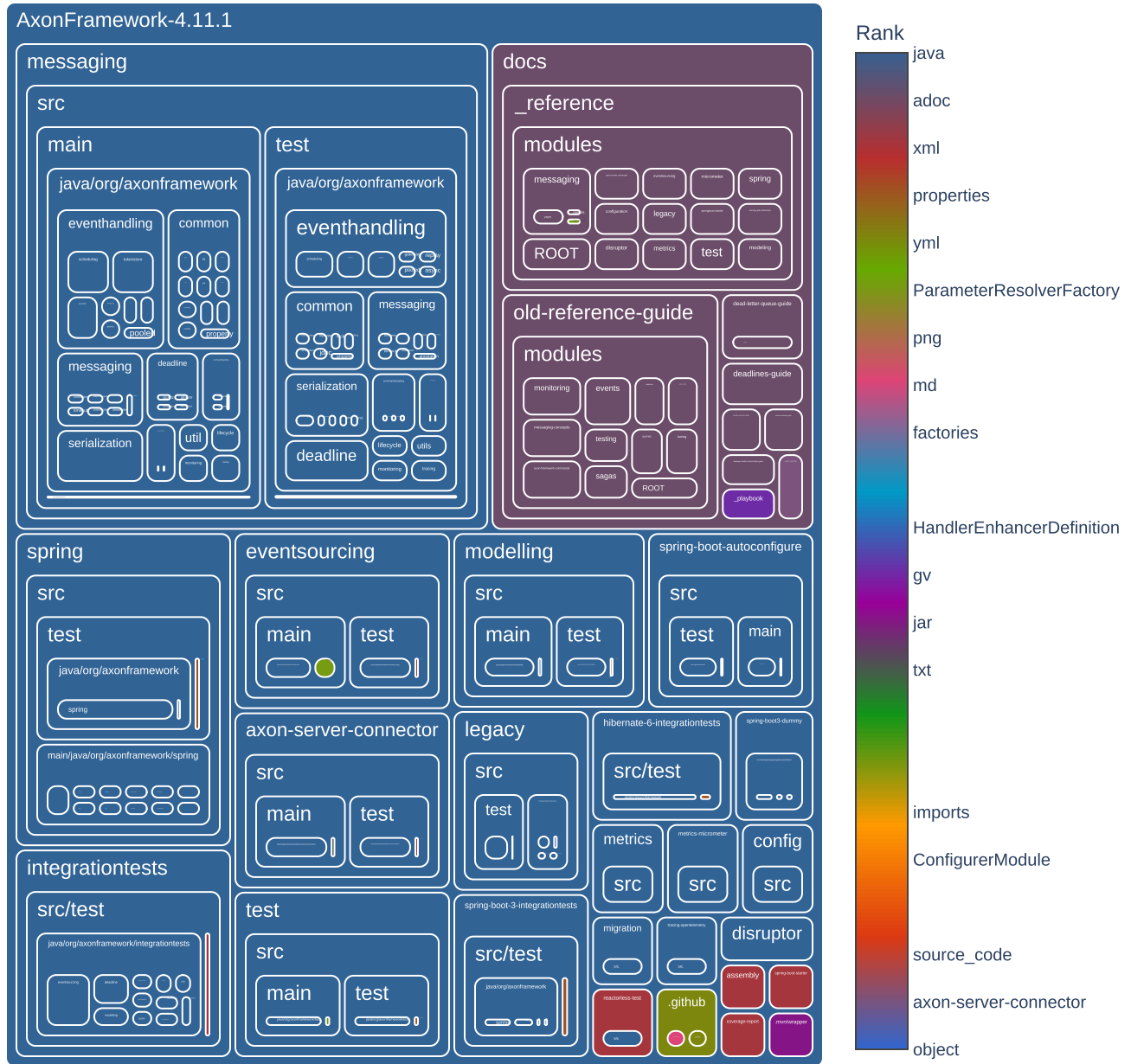
Directories and their file count



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Most frequent file extension per directory

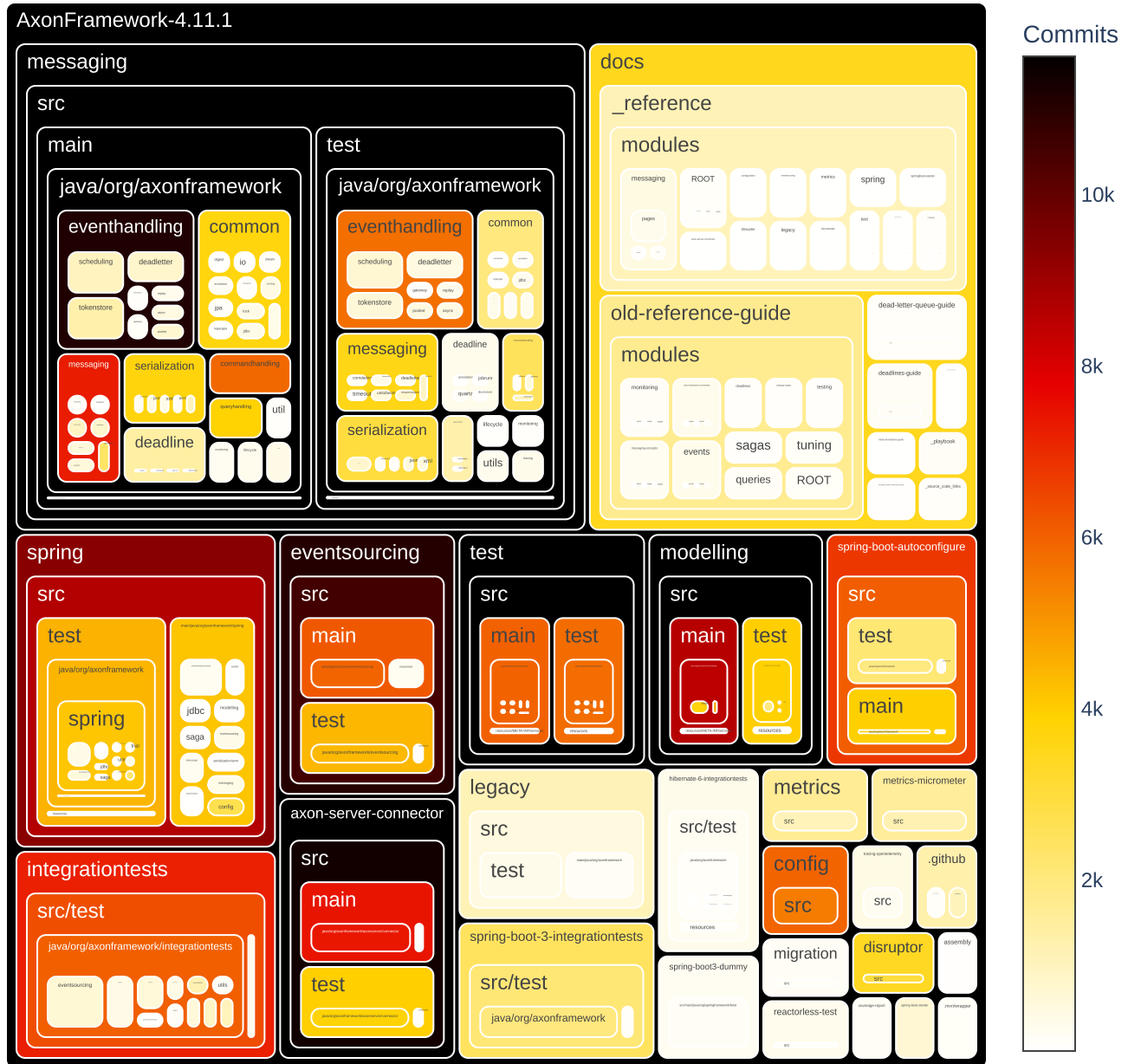
Most frequent file extension per directory



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Number of commits per directory

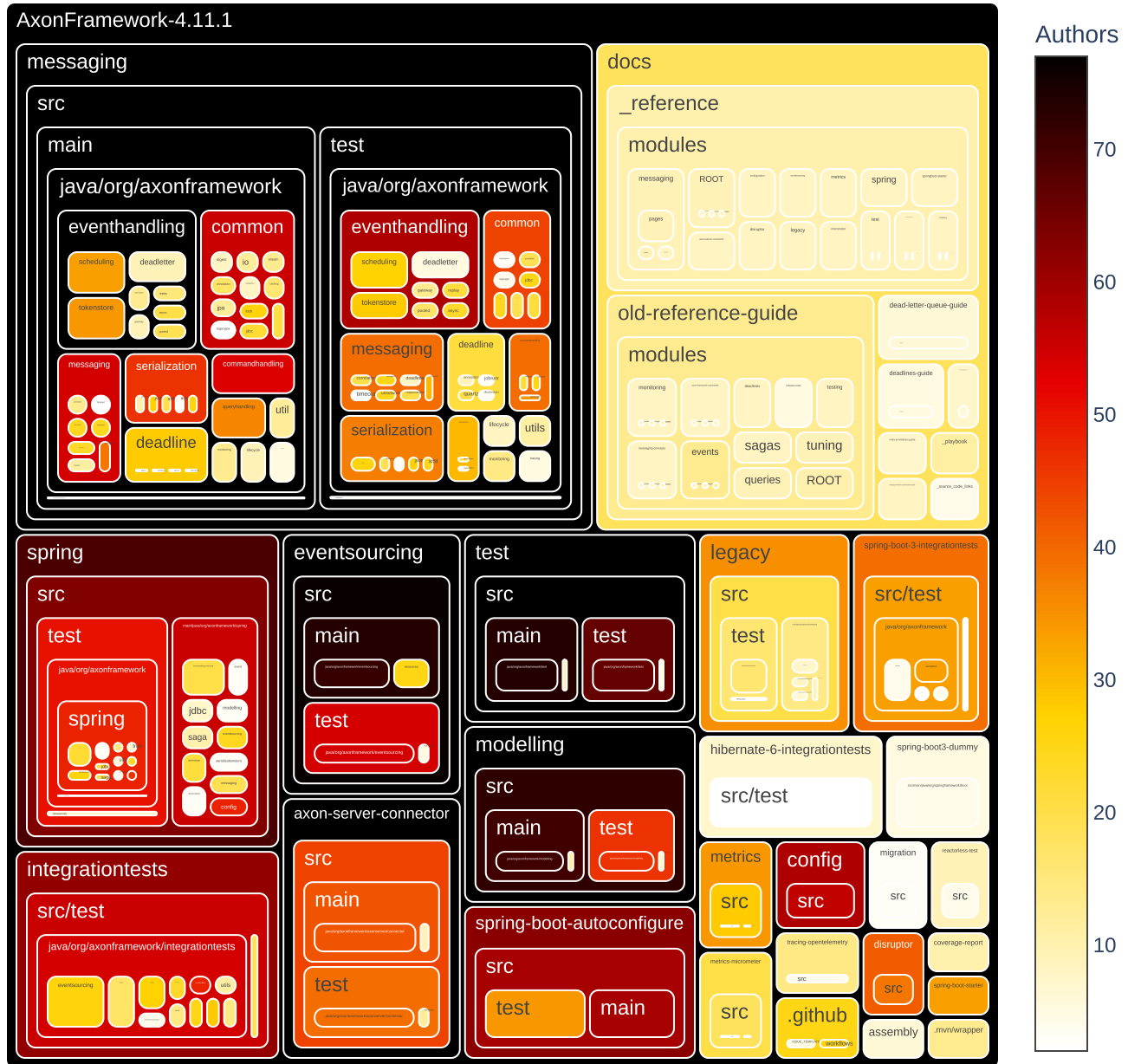
Number of git commits



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Number of distinct authors per directory

Number of distinct commit authors

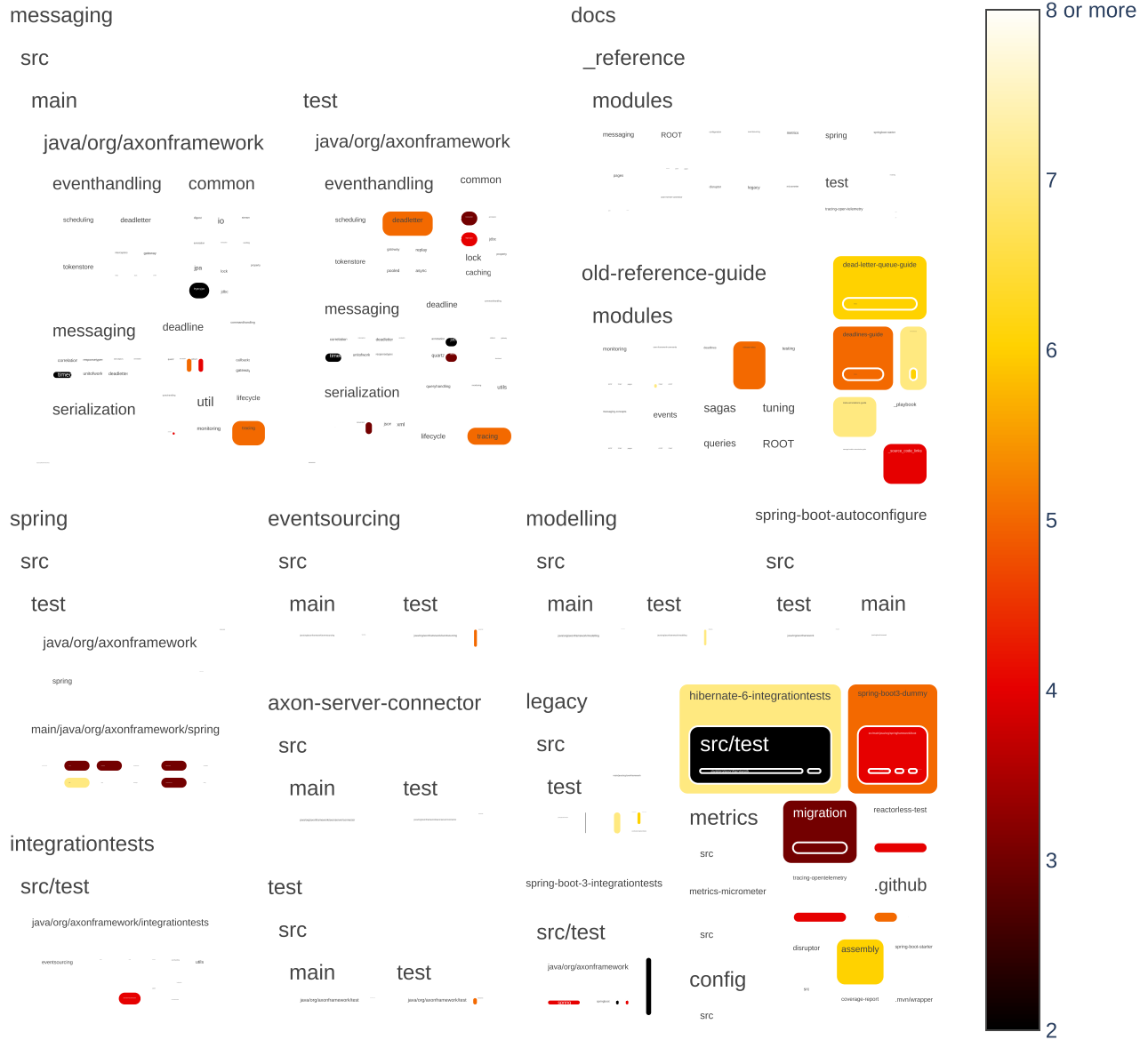


```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)

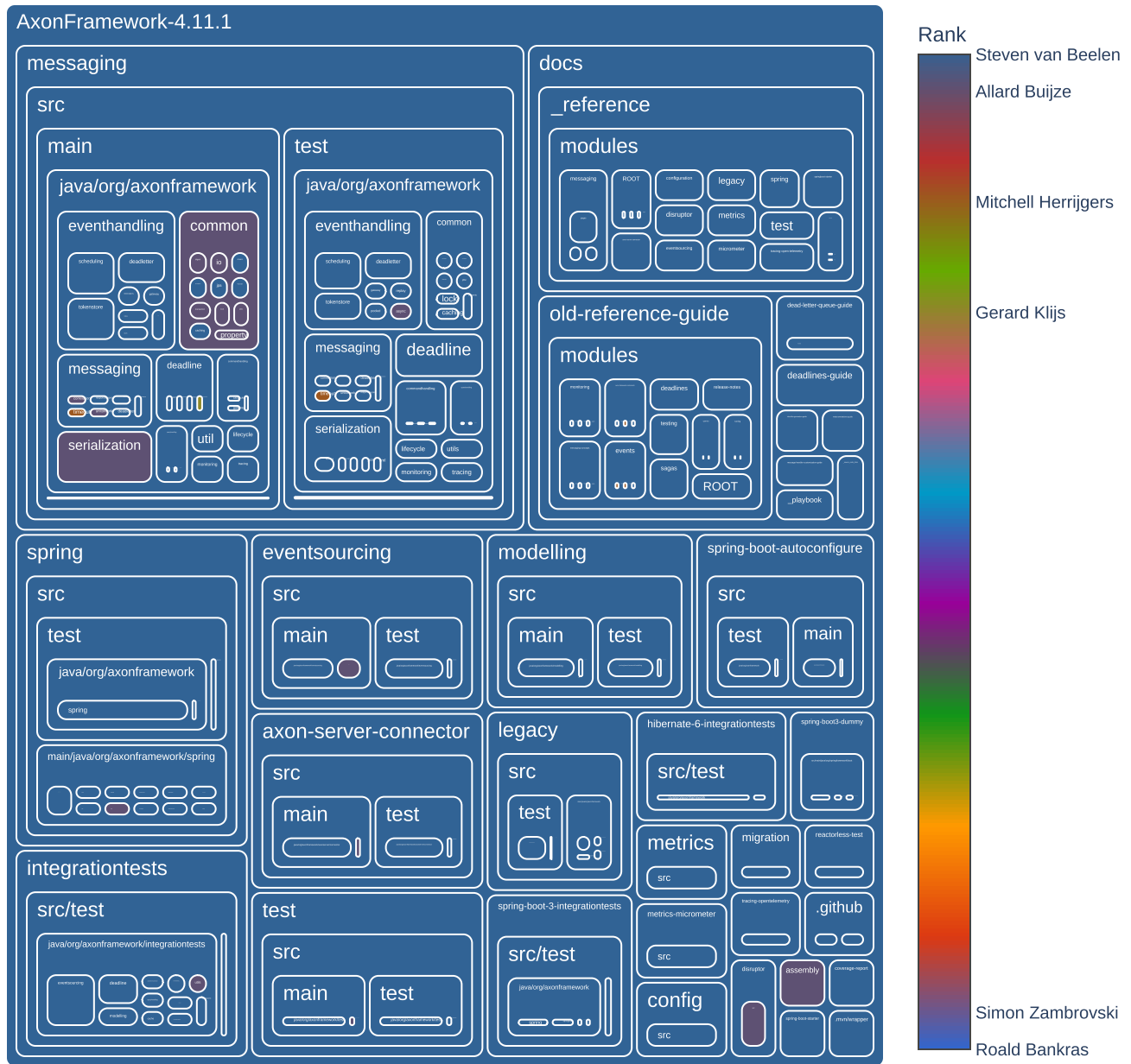
AxonFramework-4.11.1



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Main author per directory

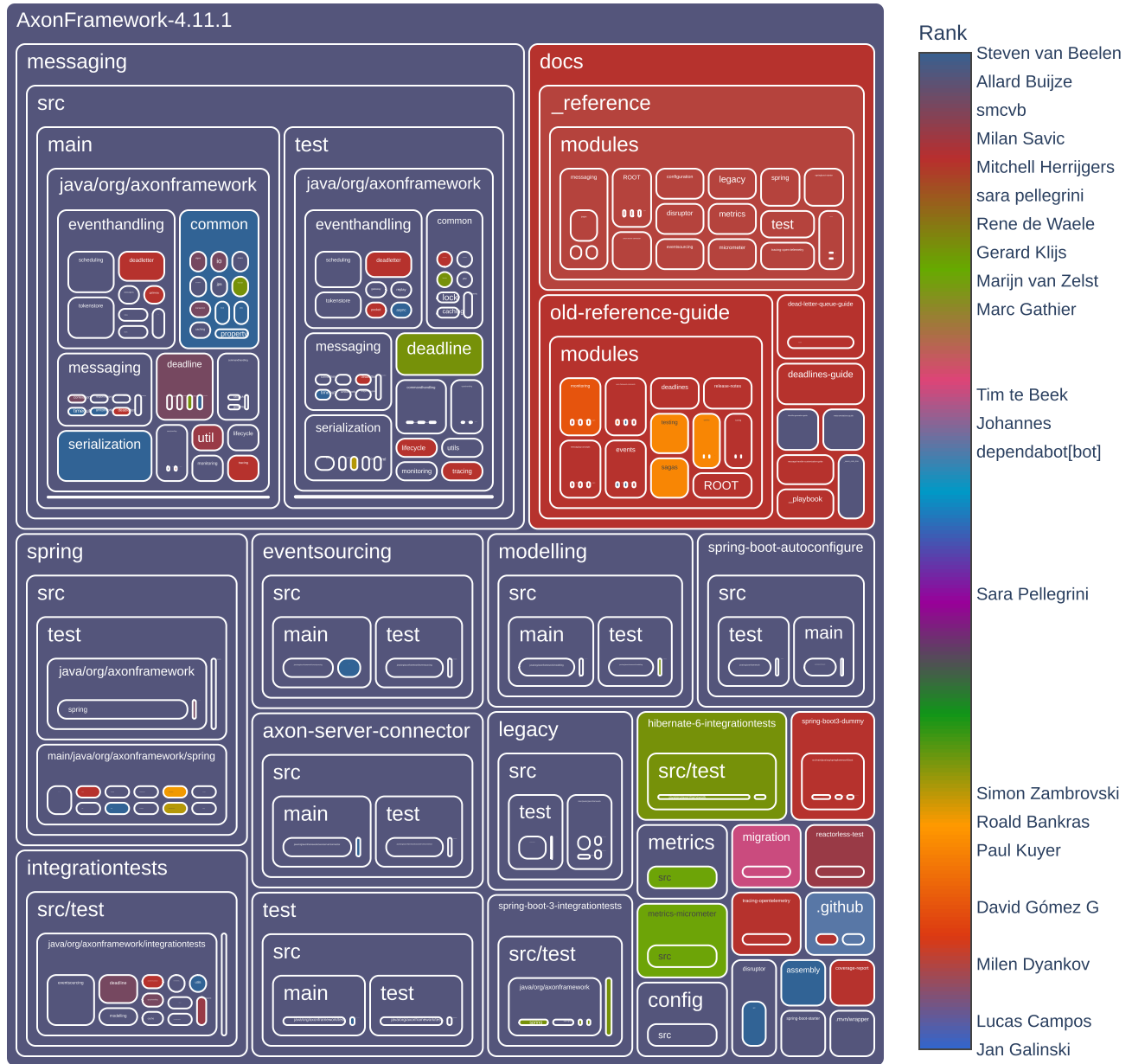
Main authors with highest number of commits



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Second author per directory

Second author with the second highest number of commits



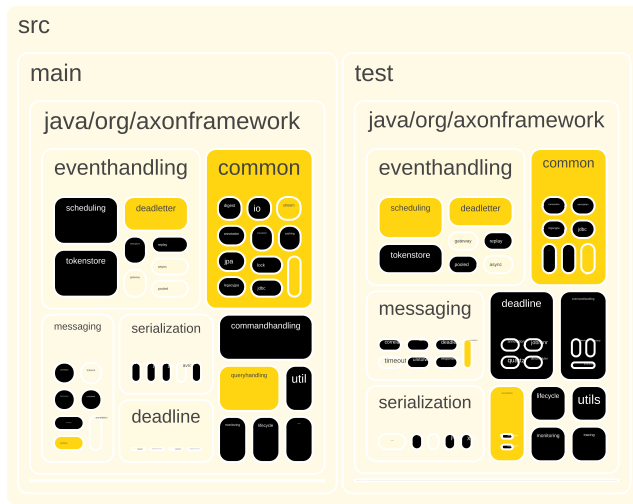
```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Days since last commit per directory

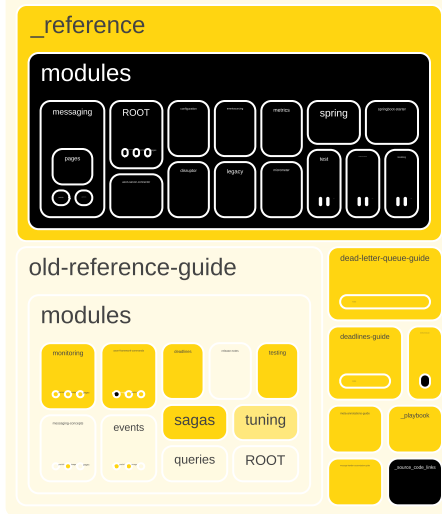
Days since last commit

AxonFramework-4.11.1

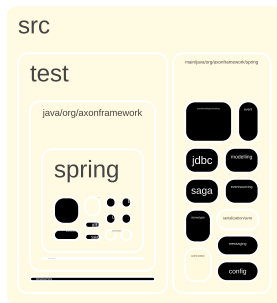
messaging



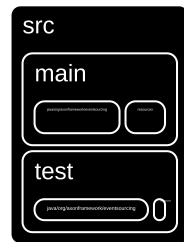
docs



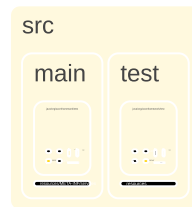
spring



eventsourcing



test



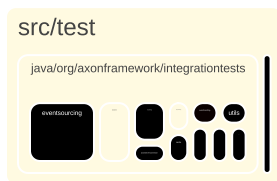
modelling



spring-boot-autoconfigure



integrationtests



axon-server-connector



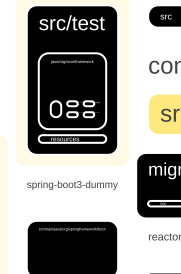
legacy



spring-boot-3-integrationtests



hibernate-6-integrationtests



metrics



metrics-micrometer



Days

250

200

150

100

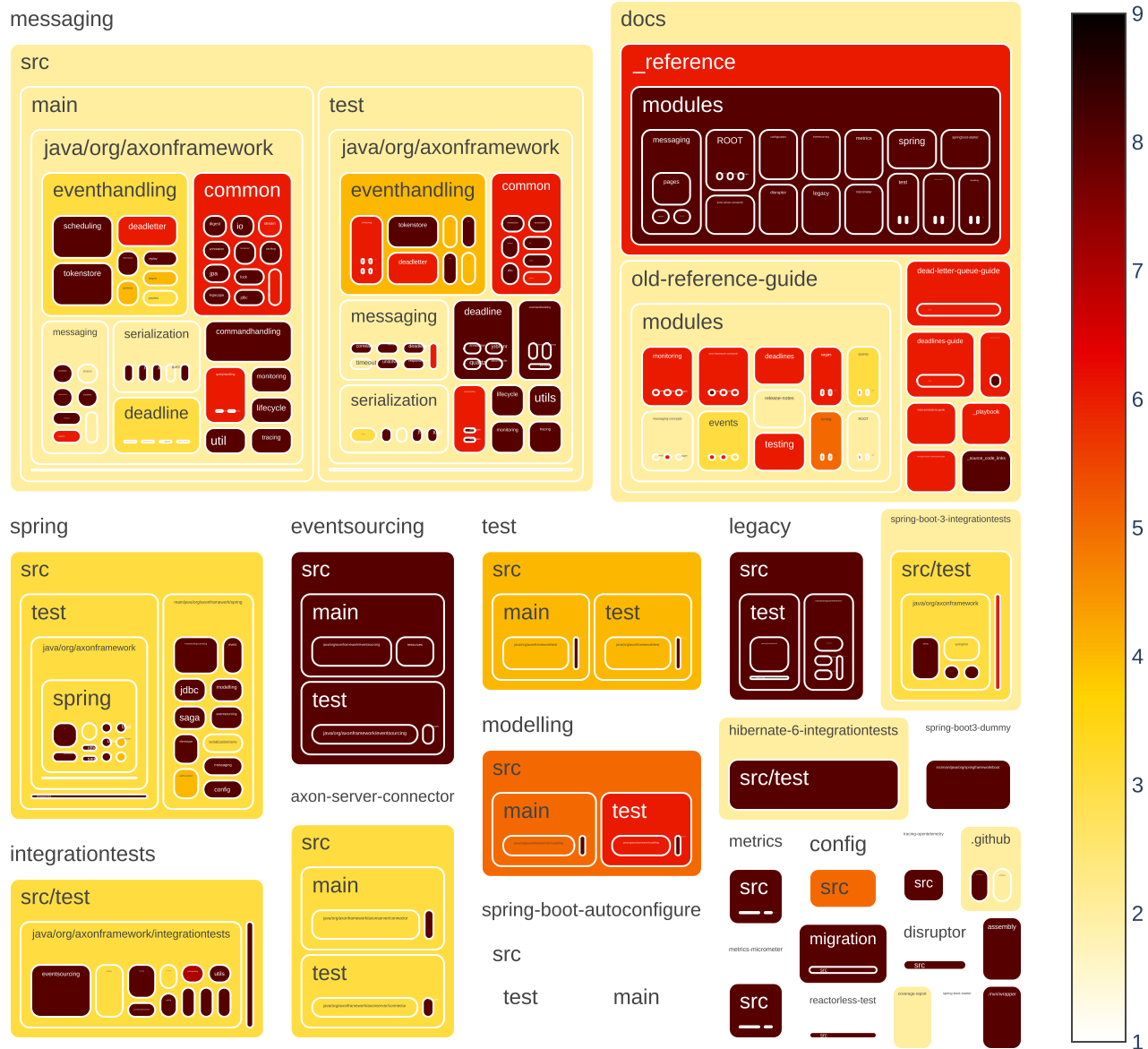
```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Days since last commit per directory (ranked)

Rank of days since last commit

AxonFramework-4.11.1

messaging



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Days since last file creation per directory

Days since last file creation

AxonFramework-4.11.1

messaging

src

main

java/org/axonframework

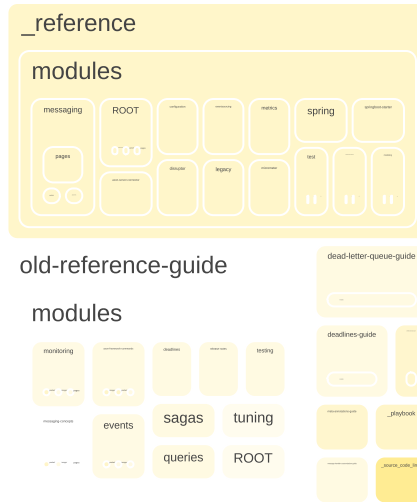


test

java/org/axonframework



docs



spring

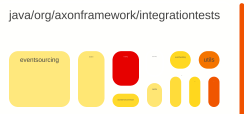
src

test



integrationtests

src/test



eventsourcing

src

main



src

main



test

src

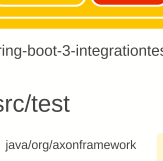
main



legacy

src

test



modelling

src

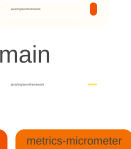
main



spring-boot-autoconfigure

src

test



src

main



Days

3500

3000

2500

2000

1500

1000

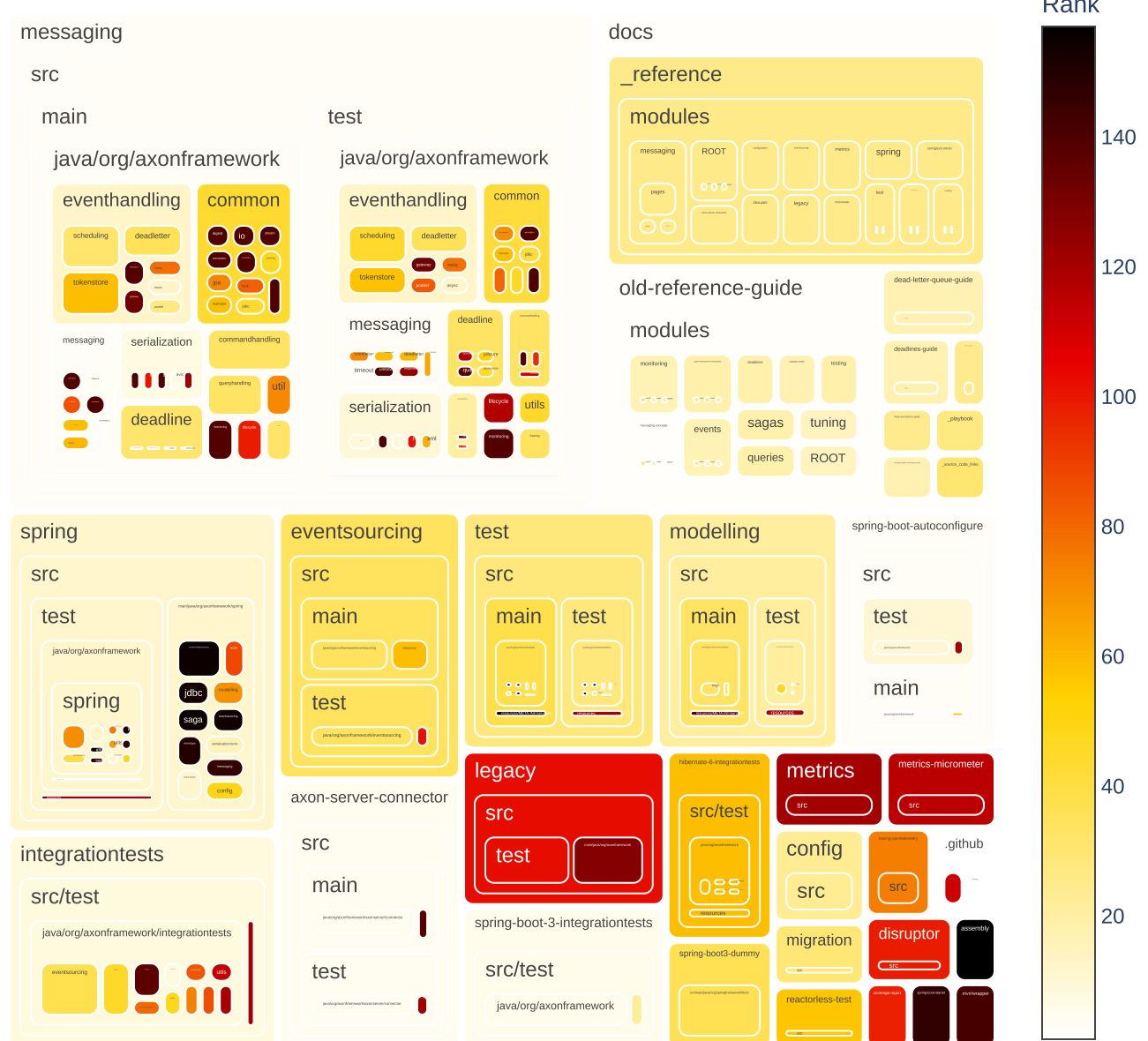
500

```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.11.1



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Days since last file modification per directory

Days since last file modification

AxonFramework-4.11.1

messaging

src

main

java/org/axonframework

eventhandling

common

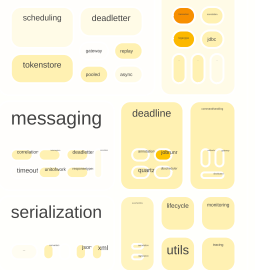


test

java/org/axonframework

eventhandling

common



docs

_reference

modules



old-reference-guide

modules



Days

2000

1500

1000

500

spring

src

test

java/org/axonframework

spring



eventsourcing

src

main

test



test

src

main

test



modelling

src

main

test



spring-boot-autoconfigure

src

test



integrationtests

src/test

java/org/axonframework/integrationtests



axon-server-connector

src

main

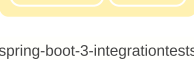
test



legacy

src

test



spring-boot-3-integrationtests

src/test

java/org/axonframework

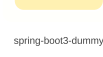


hibernate-6-integrationtests

src/test

main

test



metrics

src

config

src

migration

reactorless-test



metrics-micrometer

src

config

src

disruptor

assembly



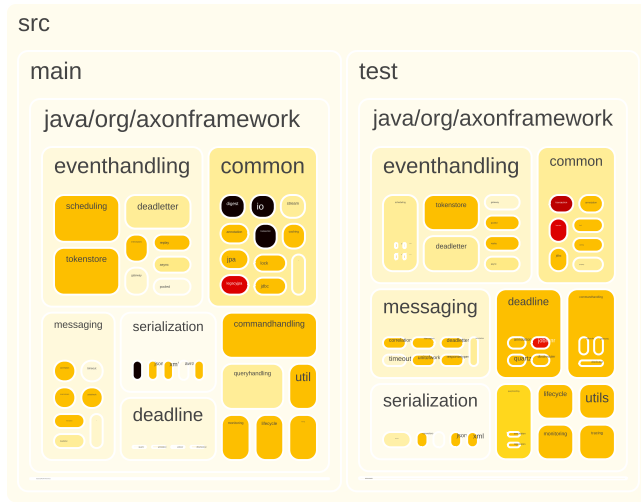
```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Days since last file modification per directory (ranked)

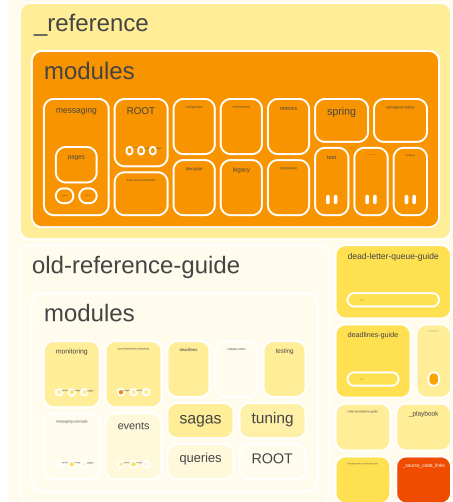
Rank of days since last file modification

AxonFramework-4.11.1

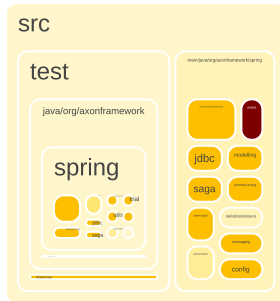
messaging



docs



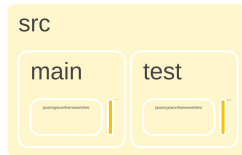
spring



eventsourcing



test



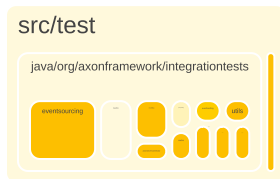
legacy



spring-boot-3-integrationtests



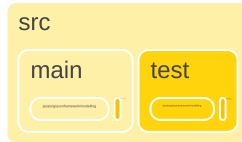
integrationtests



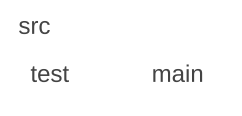
axon-server-connector



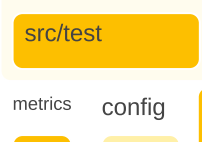
modelling



spring-boot-autoconfigure



hibernate-6-integrationtests



spring-boot-dummy



metrics



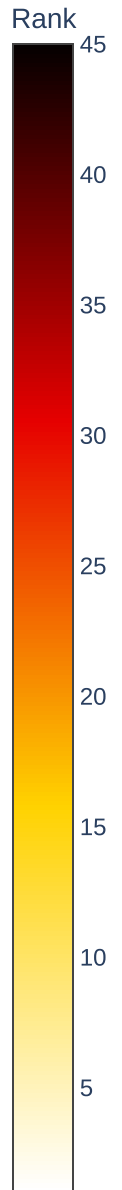
config



migration



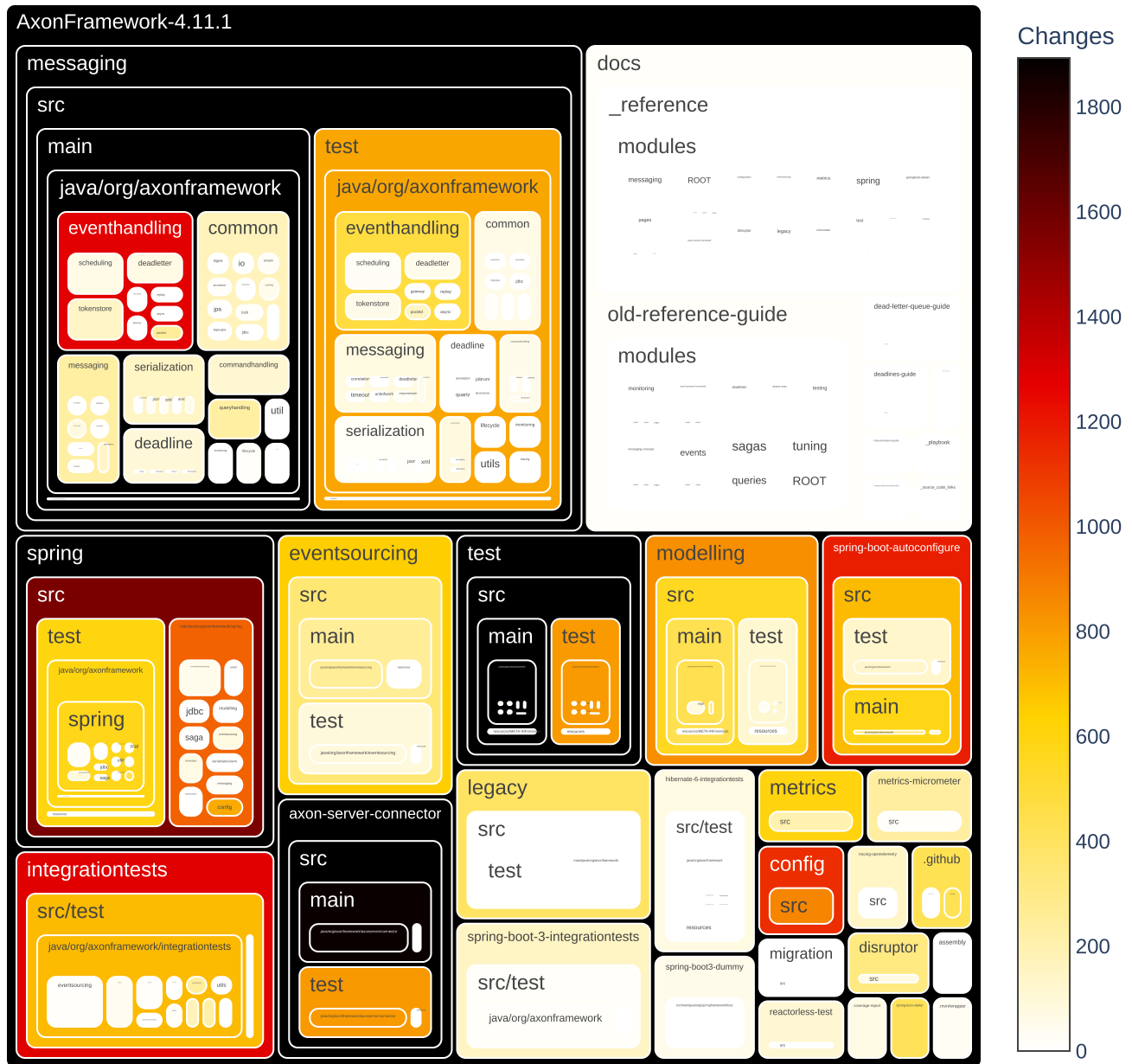
reactor-test



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

File changed frequently with other files

Pairwise file changes



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 961

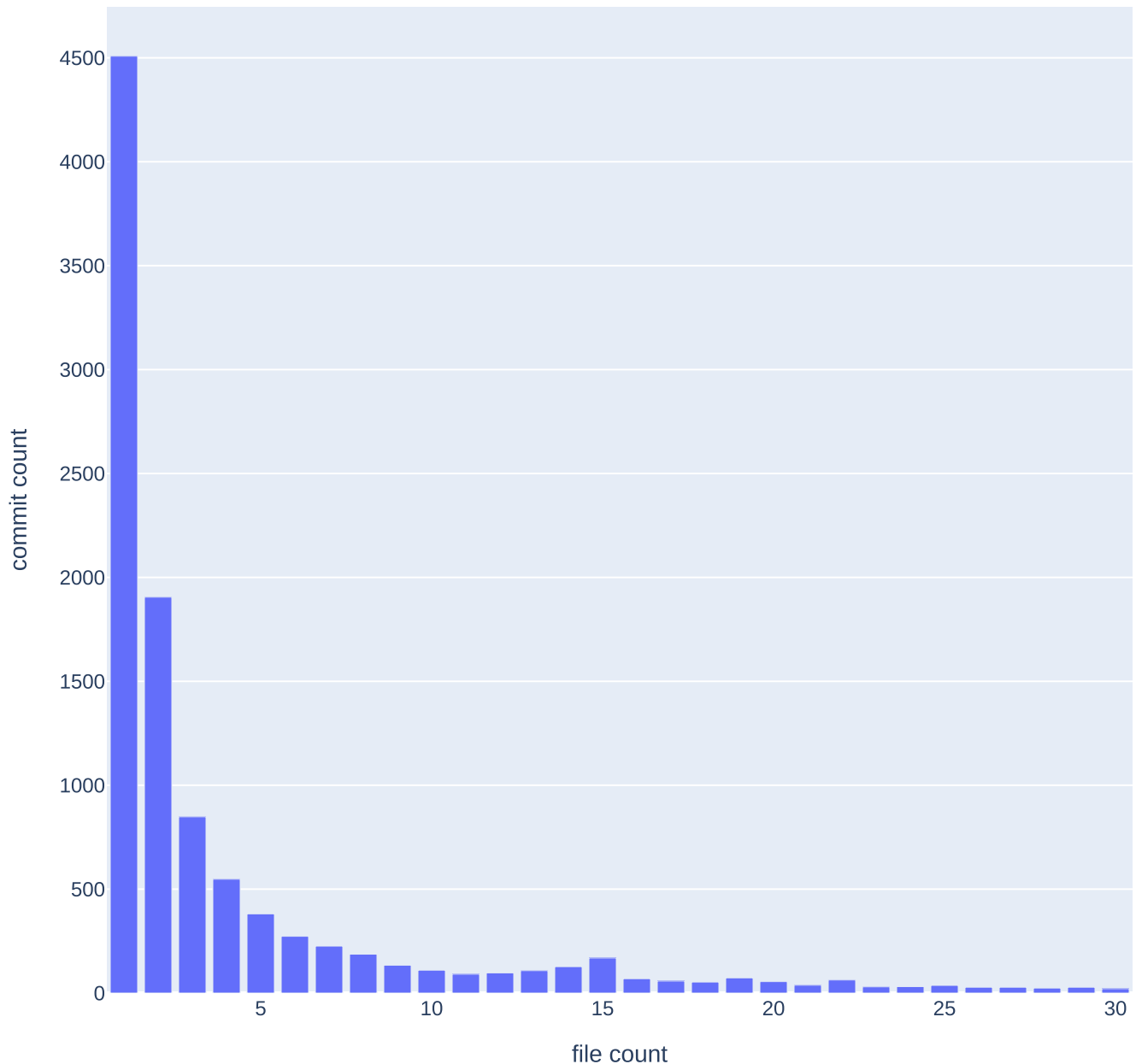
Max changed files with one commit = 2178

	filesPerCommit	commitCount
count	333.000000	333.000000
mean	344.075075	33.960961
std	440.375494	274.606293
min	1.000000	1.000000
25%	84.000000	1.000000
50%	186.000000	2.000000
75%	428.000000	5.000000
max	2178.000000	4508.000000

	filesPerCommit	commitCount
0	1	4508
1	2	1905
2	3	848
3	4	548
4	5	381
5	6	273
6	7	226
7	8	187
8	9	134
9	10	110
10	11	91
11	12	97
12	13	108
13	14	127
14	15	169
15	16	69
16	17	58
17	18	53
18	19	73
19	20	55
20	21	38
21	22	63
22	23	30
23	24	31
24	25	37
25	26	28
26	27	28
27	28	24
28	29	28
29	30	21

Bar chart with the number of files per commit distribution

Changed files per commit



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.
- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

Data Preview

```
Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: cardinality)} {position: line: 9, column: 48, offset: 576} for query:
"// List pair of files that were changed together and that have a declared dependency between each other.\n\n MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n //De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n //WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n WITH firstCodeFile.fileName AS firstFileName\n      ,secondCodeFile.fileName AS secondFileName\n      ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n      ,pairwiseChange.commitCount AS commitCount\n      ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\n RETURN dependencyWeight\n      ,commitCount\n      ,fileDistanceAsFewestChangeDirectoryCommands\n      // ,count(*) AS occurrences\n      // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\n ORDER BY dependencyWeight, commitCount\n\n // MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n // MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n // WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n // RETURN firstCodeFile.fileName AS firstFileName\n //      ,secondCodeFile.fileName AS secondFileName\n //      ,dependency.weight AS dependencyWeight\n //      ,pairwiseChange.commitCount AS commitCount\n // ORDER BY dependencyWeight, commitCount\n\n // MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n // WITH count(DISTINCT relation) AS relatedFilesCount\n //      ,collect(DISTINCT relation) AS relations\n // UNWIND relations AS relation\n // WITH relatedFilesCount\n //      ,coalesce(relation.commitCount, 0) AS commitCount\n //      ,coalesce(relation.weight, 0) AS dependencyWeight\n //      ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n // RETURN dependencyWeight\n //      ,commitCount\n //      ,fileDistanceAsFewestChangeDirectoryCommands\n // ORDER BY dependencyWeight, commitCount"
```

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
0	1	12	0
1	1	12	0
2	1	12	4
3	1	12	0
4	1	12	0
5	1	12	4
6	1	12	0
7	1	12	2
8	1	12	0
9	1	12	4
10	1	13	0
11	1	13	4
12	1	13	0
13	1	13	4
14	1	14	4
15	1	14	2
16	1	15	0
17	1	15	2
18	1	15	0
19	1	16	0

Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
count	166.000000	166.000000	166.000000
mean	9.590361	25.204819	1.373494
std	19.230680	16.573260	1.570260
min	1.000000	12.000000	0.000000
25%	1.250000	14.250000	0.000000
50%	3.000000	19.000000	0.000000
75%	10.000000	28.000000	2.000000
max	154.000000	95.000000	4.000000

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.200368	-0.141013
commitCount	0.200368	1.000000	-0.145947
fileDistanceAsFewestChangeDirectoryCommands	-0.141013	-0.145947	1.000000

'Pairwise changed git files compared to dependency weights - Spearman Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.214529	-0.091448
commitCount	0.214529	1.000000	-0.133180
fileDistanceAsFewestChangeDirectoryCommands	-0.091448	-0.133180	1.000000

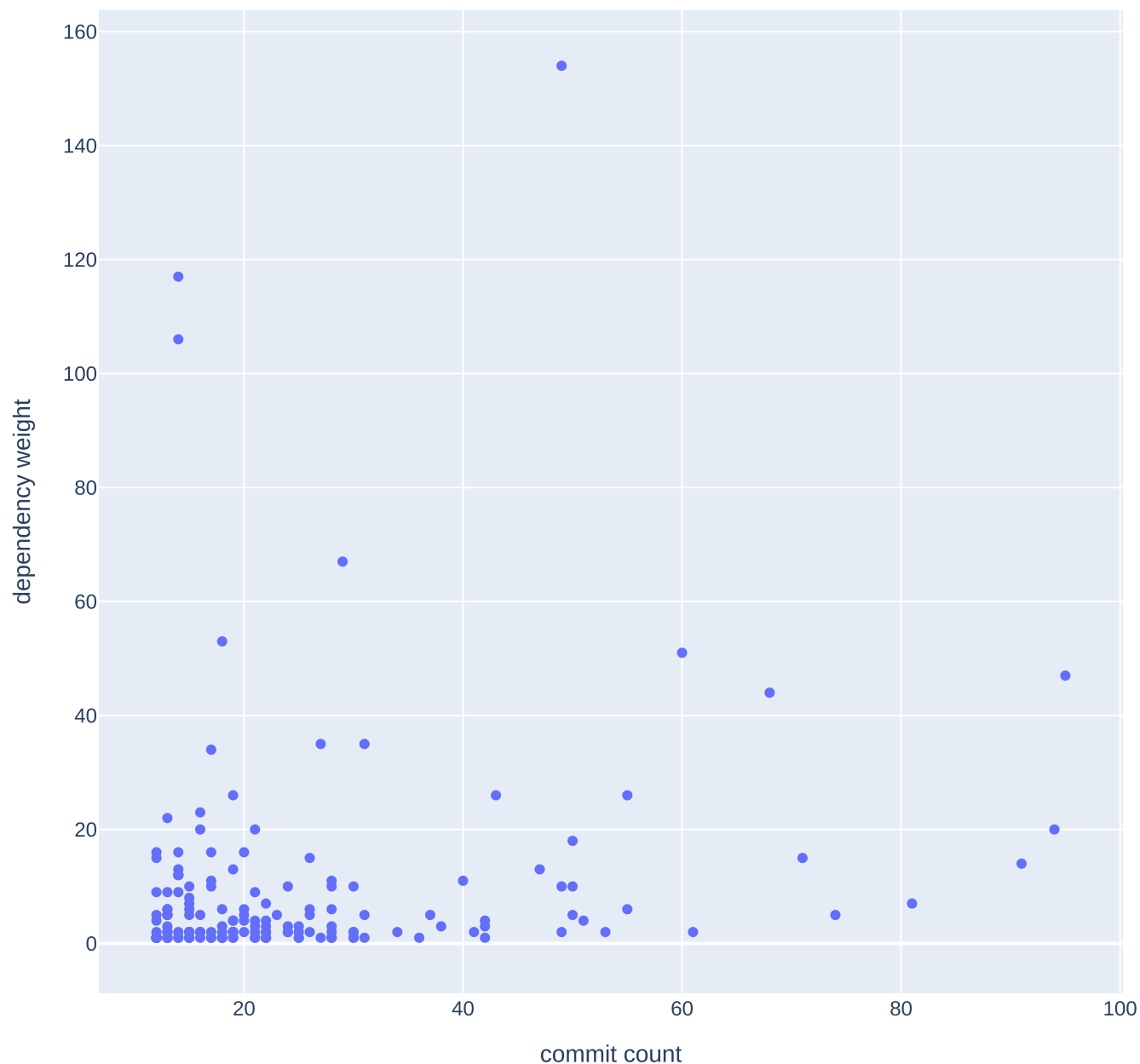
'Pearson Correlation with p-value for commitCount and dependencyWeight'

PearsonRResult(statistic=0.20036758755264605, pvalue=0.009643647845018762)

'Spearman Correlation with p-value for commitCount and dependencyWeight'

SignificanceResult(statistic=0.21452881933918705, pvalue=0.005511439972198768)

Pairwise changed files: Number of changes (commitCount) vs. dependency weight



```
os.path.abspath=/home/runner/work/code-graph-analysis-examples/code-graph-analysis-examples/jupyter
os.path.curdir=.
```

WordCloud of git authors

	word	frequency
0	Steven van Beelen	3360
1	Allard Buijze	2721
2	smcvb	822
3	Milan Savic	449
4	Mitchell Herrijgers	309
5	Gerard Klijs	246
6	Marc Gathier	231
7	sara pellegrini	163
8	Rene de Waele	121
9	Jетро Coenradie	117

Wordcloud of git authors

